# Can Movie Posters Predict Success and Ratings?

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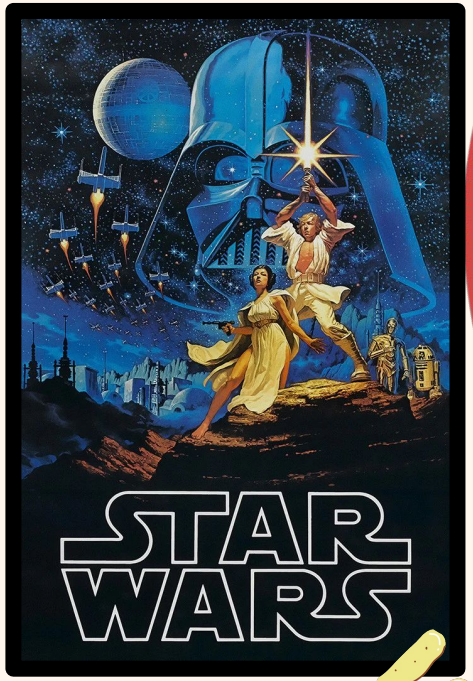
# Motivation

**Context**

- Industry is worth 112 billion USD
- 10-15 major genres, 100+ subgenres
- Movies need 2.5 times its production budget to reach breakeven

**Research Interest**

- Interest in Movies/Cinema
- What makes a good movie poster?
- Tool to predict success?



# Project Goal

**Goal**

- Classify movie attributes such as genre and box office results based on poster images

**Research Question**

- To what degree can computer models accurately classify movie genres, ratings, and box office performance from poster images?

**Modeling Approach**

- Fine-tuned ResNet-50 model predicting genre, ratings, and revenue from posters vs a baseline adjusted for inflation

# Data Acquisition

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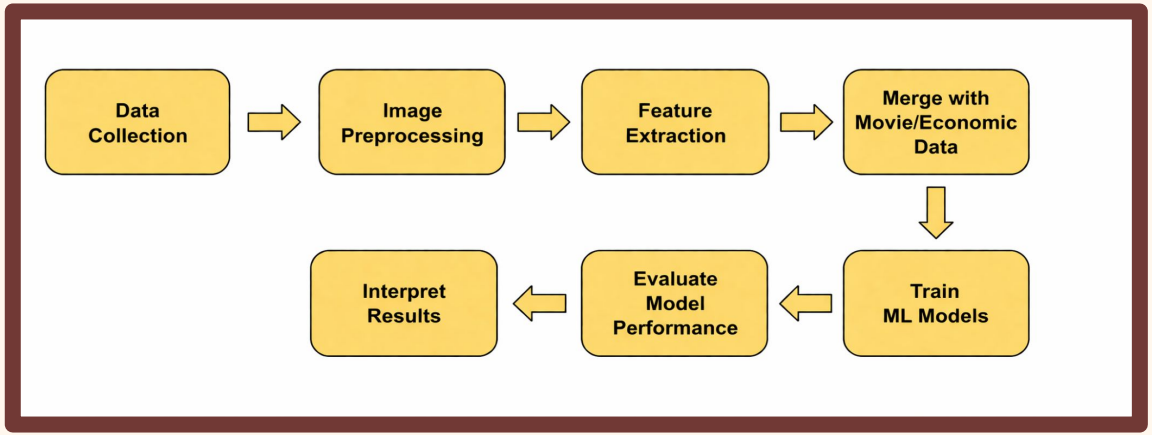
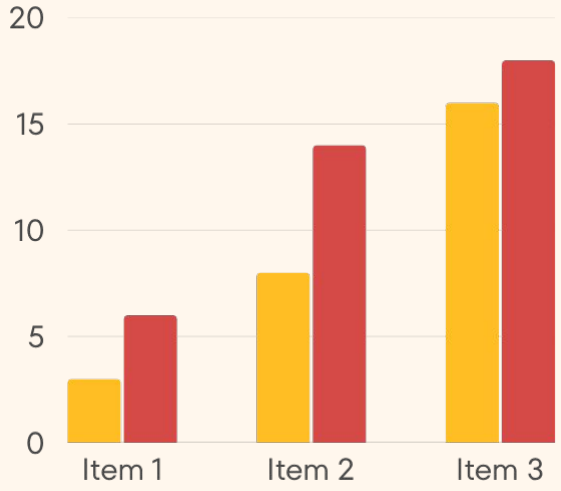
# Data Overview

# Analysis Plan + Justification



# ANALYSIS

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# Analysis Decision

**The Decision**

- Use inflation-adjusted box office revenue to ensure fair comparisons between old and new movies

**Why it was tricky**

- Revenue spans decades, nominal figures influenced by inflation. Older films appear less successful due to lower price levels, making it hard to track patterns over time.

**The Result**

- Allowed for consistent performance metrics. Improved models ability to learn relationship between poster features and financial success.

# Conclusion & Results

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# Next Steps

## New lines of Exploration

- Extract and analyze key frames from movie trailers
- Use facial recognition to account for star actors

## Improvements

- Test higher resolution images
- Apply more techniques to account for genre imbalance
- Benchmark against a more robust financial baseline.

## New Questions

- Does a poster's predictive power change across decades?
- Do specific color palettes universally boost audience ratings?

# References


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[2] “Consumer Price Index (CPI) Databases,” U.S. Bureau of Labor Statistics, https://www.bls.gov/data/home.htm#prices (accessed Apr. 1, 2026).

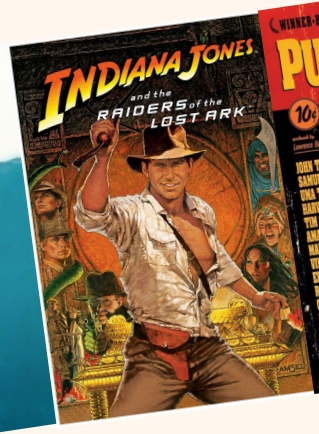
[3] “Movie Dataset Financials,” Google Sheets, https://docs.google.com/spreadsheets/d/1bYVyxLtbq9bBDxdLtSXQOpZlqNKHLGtO/ (accessed Apr. 1, 2026).

[4] “Getting Started,” TMDB Developer Documentation, https://developer.themoviedb.org/docs/getting-started (accessed Apr. 1, 2026).

[5] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770-778. [Online]. Available: https://arxiv.org/abs/1512.03385 (accessed Apr. 1, 2026).

# THANK YOU!

JAWS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RAIDERS of the
LOST ARK

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PULP FICTION
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JOHN TRAVOLTA
SAMUEL L. JACKSON
UMA THURMAN
HARVEY KEITEL
TIM ROTH
AMANDA PLUMMER
MARIA de MEDEIROS
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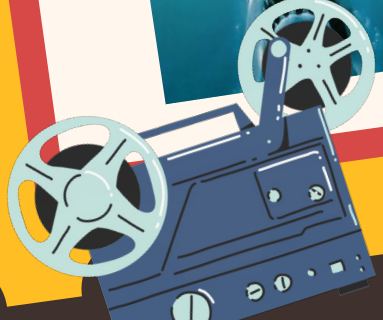
ROSANNA ARQUETTE
CHRISTOPHER WALKEN
BRUCE WILLIS

LAWRENCE OF ARABIA

JIM CARREY



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